Gesture Recognition with the Nano 33 BLE Sense

Learn how to use the built in gesture sensor of the Nano 33 BLE Sense to control the built in RGB LED.

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In this tutorial we will use an Arduino Nano 33 BLE Sense for gesture recognition, made possible by the embedded **APDS9960** sensor.

We will use the sensor to print out simple hand gesture directions and control the board's RGB LED accordingly. In addition, we will program our board to blink the built-in LED and change colors to the RGB LED. according to the direction of the set gestures. The code will read simple **Up-Down-Right-Left** hand motions.

Goals

The goals of this project are:

- Learn what a APDS9960 sensor is.
- Use the APDS9960 library.

Help

Arduino Nano 33 BLE Sense.

Create your own gesture detection monitor.

Learn how to control the built-in LED and RGB LED, through hand gestures.

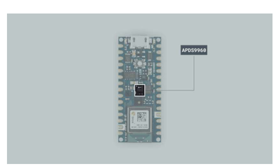
Hardware & Software Needed

This project uses no external sensors or components.

In this tutorial we will use the **Arduino Create Cloud Editor** to program the board.

APDS9960 Sensor

The APDS9960 sensor is a multipurpose device that features advanced gesture detection, proximity detection, Digital Ambient Light Sense (ALS) and Color Sense (RGBC).



The APDSS9960 sensor.

The sensor's gesture detection utilizes four directional photodiodes to sense reflected IR energy (sourced by the integrated LED) to convert physical motion information (i.e.

It features:

- Four separate diodes sensitive to different directions.
- Ambient light rejection.
- Offset compensation.
- Programmable driver for IR LED current.
- 32 dataset storage FIFO.
- Interrupt driven I2C-bus communication.

If you want to read more about the APDS9960 sensor module see here.

The Library

The APDS9960 library allows us to use the sensor available on the board, to read gestures, color, light intensity and proximity. The library includes some of the following functions:

```
begin()
end()
gestureAvailable()
readGesture()
colorAvailable()
readColor()
proximityAvailable()
readProximity()
setGestureSensitivity
setInterruptPin()
setLEDBoost()
```

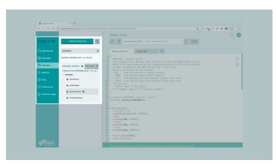
If you want a deeper knowledge on any of the functions of the library, you can check the Arduino reference for this library.

For the purposes of this tutorial we will only focus on the gesture readings, which are based on the detection of the movement of the hand over four photodiodes inside the sensor.

Creating the Program

1. Setting up

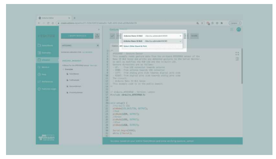
Let's start by opening the Arduino Cloud Editor, click on the **Libraries** tab and search for the **APDS9960** library. Then in **> Examples**, open the **GestureSensor** sketch and once it opens, you could rename it as Gesture&LEDs.



Finding the library in the Cloud Editor.

2. Connecting the board

Now, connect the Arduino Nano 33 BLE Sense to the computer and make sure that the Cloud Editor recognizes it, if so, the board and port should appear as shown in the image below. If they don't appear, follow the instructions to install the plugin that will allow the Editor to recognize your board.



Selecting the board.

3. Blink patterns according to hand gestures

Now we will need to modify the code on the example, in order to change the color of the RGB LED and the built-in LED respectively according to the direction of our hand gesture.

After including the `Arduino_APDS9960.h` library, we will need to configure the specified pins (22, 23, 24, LED_BUILTIN) at the beginning of the `setup()` section, to behave as output:

```
//in-built LED
pinMode(LED_BUILTIN, O
//Red
pinMode(LEDR, OUTPUT);
//Green
pinMode(LEDG, OUTPUT);
//Blue
pinMode(LEDB, OUTPUT);
```

and then at the end, we need to turn all the LEDs OFF by adding the following statements

```
// Turining OFF the RG
  digitalWrite(LEDR, H
  digitalWrite(LEDG, H
  digitalWrite(LEDB, H
```

In the `loop()` section the `if()` statement is checking that the gesture sensor is available and if it is, it reads for any incoming gesture detection.

Next, in the `switch()` statement we will add different actions to be performed according to conditions, in this case the direction of the hand gesture will define those conditions.

If the sensor detects motions (up, down, left or right) we can add the following code snippets between the **Serial.println()** and the **break;** in each switch case, to activate the red, blue and green colors of the RGB LED and the orange on the built-in LED.

In the `GESTURE_UP` case, the RGB LED will glow **red** for a second:

```
digitalWrite(LEDR, LOW
delay(1000);
digitalWrite(LEDR, HIG
```

In the `GESTURE_DOWN` case, the RGB LED will glow **green** for one

```
1 digitalWrite(LEDG, LOW
2 delay(1000);
3 digitalWrite(LEDG, HIG
```

In the `GESTURE_LEFT` case, the RGB LED will glow **blue** for one second:

```
1 digitalWrite(LEDB, LOW
2 delay(1000);
3 digitalWrite(LEDB, HIG
```

Lastly, in the `GESTURE_RIGHT` case, the small built-in LED will glow **orange** for one second:

```
1 digitalWrite(LED_BUILT
2 delay(1000);
3 digitalWrite(LED_BUILT
```

Now the code is complete!

4. Complete code

```
/*
  APDS9960 - Gesture
  This example reads
  Nano 33 BLE Sense
  Gesture directions
  - UP:    from USB
  - DOWN:  from ante
  - LEFT:  from anal
  - RIGHT: from dig
  The circuit:
  - Arduino Nano 33
  This example code
*/

#include <Arduino_A

void setup() {
  Serial.begin(9600)
  //in-built LED
  pinMode(LED_BUILTI
  //Red
  pinMode(LEDR, OUTP
  //Green
  pinMode(LEDG, OUTP
  //Blue
  pinMode(LEDB, OUTP

  while (!Serial);
```

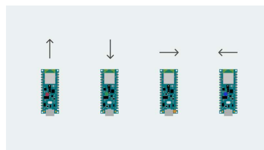
Testing It Out

Triggering the gesture sensor.

After you have successfully verified and uploaded the sketch to the board, open the Serial Monitor from the menu on the left.

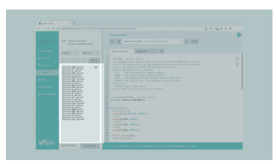
In order to test out the code,

position in front of you (USB port facing down) and carry on by making directional UP-DOWN-RIGHT-LEFT hand gestures. Try to make your movement as clear as possible, yet subtle enough for the sensor to pick it up.



Direction of hand gestures.

Here is a screenshot example of the sketch returning values.



Gesture detections printed in the Serial Monitor.

Troubleshoot

Sometimes errors occur, if the code is not working there are some common issues we can troubleshoot:

- Missing a bracket or a semicolon
- Arduino board connected to the wrong port
- Accidental interruption of cable connection

Conclusion

In this tutorial we learned what a **APDS9960** sensor is, how to use the one embedded in the Arduino Nano 33 BLE Sense board and the APDS9960 library, as well as how to create your own gesture detection monitor which can operate the RGB and built-in LED in various color patterns.